

Consumer Spending



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INTRODUCTION:

Good evening, class. My name is Mina. Please welcome my teammate, Kiwaski. We analyzed Consumer Spending to provide valuable insights into customer demographics, income, spending habits, and purchase behavior. This focus on customer behavior is particularly relevant to our finance and data analysis roles.

Objective:

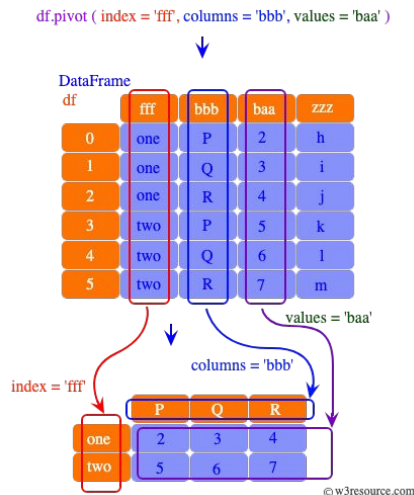
We used a public dataset from Kaggle to analyze the data. The CSV file contains customer data with various attributes related to demographics, income, spending habits, and purchase behavior. Each row in the file represents a different customer, while the columns represent different features. Thanks to Aditya Goyal (Owner) and Kaggle for providing the dataset.

Overview

We used Pandas and Tableau for this project as the technologies, in addition of machine learning to predict that the United States consumer spending will exceed that of other countries worldwide.

The analysis begins by cleaning, transforming, and preparing the dataset for analysis using Pandas, followed by creating interactive visualizations in Tableau to uncover insights.

We also implemented machine learning models to predict consumer spending trends using linear regression, decision trees, and random forests. The goal is to identify the most effective model for predicting future spending based on historical data.



Creation of the Consumer Spending

```
In [34]: # Display the number of rows in the DataFrame
print("Rows in DataFrame: ", len(df_customer_data))
```

```
# Display the first 10 rows of the DataFrame
df_customer_data.head(10)
```

Rows in DataFrame: 1000

```
Out[34]:
```

	name	age	gender	education	income	country	purchase_frequency	spending
0	Teresa Williams MD	42	Female	High School	53936	Slovenia	0.9	13227.120
1	Christine Myers	49	Female	Master	82468	Aruba	0.6	12674.040
2	Dwayne Moreno	55	Male	Bachelor	56941	Cyprus	0.3	5354.115
3	Amy Norton	24	Female	Bachelor	60651	Palau	0.2	2606.510
4	Tonya Adams	64	Male	Master	81884	Zambia	0.9	18984.780
5	Charles Smith	24	Female	PhD	61444	Libyan Arab Jamahiriya	1.0	18072.200
6	Misty Moody	58	Male	PhD	78024	Italy	0.9	21511.080
7	Brian Kent	34	Male	High School	32225	Pakistan	0.7	6727.875
8	Stacie Ray	63	Male	PhD	63664	Venezuela	0.4	9273.280
9	Brian Townsend	64	Female	High School	45043	San Marino	1.0	16252.150

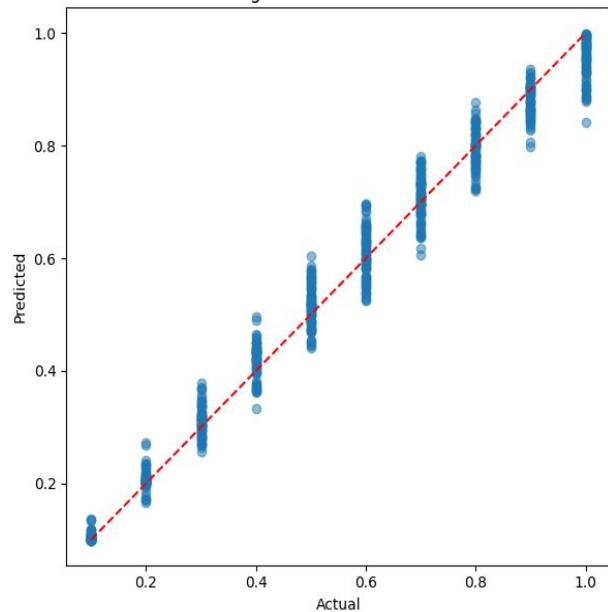
```
''' [34]: # Drop the non-beneficial ID columns, 'education', 'country'
df_customer_data.drop(columns = ["education", "country"],inplace =True)
df_customer_data.head()
#Convert the column to float
df_customer_data = df_customer_data.astype(float)

#Print the DataFrame
print(df_customer_data)
```

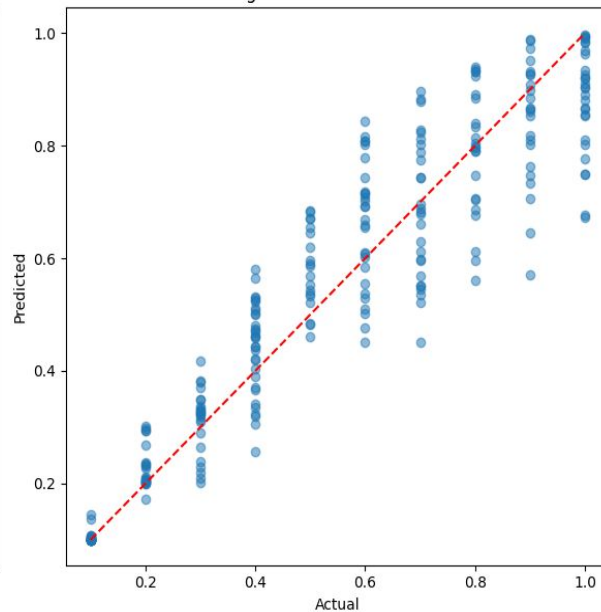
	age	income	purchase_frequency	spending
0	42.0	53936.0	0.9	13227.120
1	49.0	82468.0	0.6	12674.040
2	55.0	56941.0	0.3	5354.115
3	24.0	60651.0	0.2	2606.510
4	64.0	81884.0	0.9	18984.780
...	...	...	...	...
995	42.0	98170.0	0.7	17435.950
996	39.0	59174.0	0.9	11662.830
997	50.0	59808.0	0.4	7196.160
998	50.0	58272.0	0.7	13939.520
999	38.0	32512.0	0.5	8312.800

1000 rows x 4 columns

Training Data: Actual vs Predicted



Testing Data: Actual vs Predicted



	Metric	Train	Test
0	MSE	0.001453	0.010891
1	RMSE	0.038113	0.104362
2	R^2	0.982005	0.866624

Part II: Tableau

https://public.tableau.com/viz/Project4_17235712325600/CustomCount?:language=en-US&publish=yes&sid=&redirect=auth&displaycount=n&origin=viz_share_link



CONCLUSION

By generating summary statistics on age, income, purchase frequency, and customers' spending habits, we noticed that in 2023, BCG found that people aged 50–70 in 12 countries, including the US and China, were responsible for 27% of spending. The United Nations estimates that people over 50 spend about \$15 trillion globally, which is expected to continue increasing.



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- As of 2018, baby boomers still outspent millennials and accounted for more than half of all consumer spending in the US.

Bloomberg estimates that in 2021, Gen Z had \$360 billion in buying power, which will likely increase as more Gen Zers enter the workforce.

(born between 1946 and 1964)



The End

A group of people are silhouetted against a large window, looking out at a city skyline. The most prominent building in the background is St. Paul's Cathedral, with its large dome and classical architecture. Other buildings of varying heights and styles fill the rest of the view. The scene is dimly lit, with the primary light source being the natural light from the window, which creates the silhouettes of the people in the foreground.

Q & A